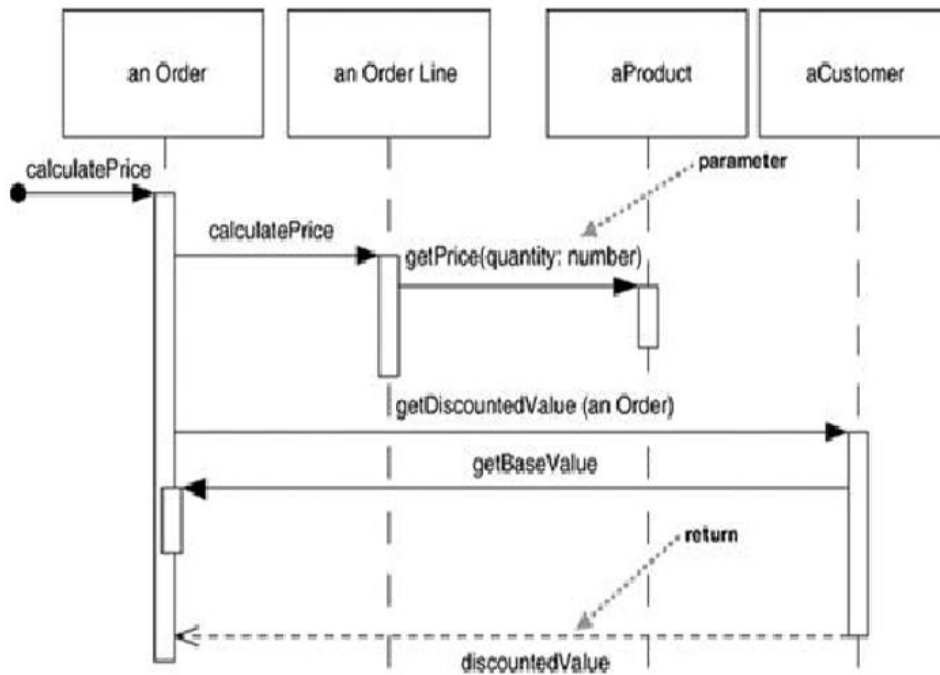


HW1: (1 倍分數)

請把附圖的 sample code 寫出來

Figure 4.2. A sequence diagram for distributed control



A:

```
class Order {
    private OrderLine orderLine;
    private Customer customer;

    public double calculatePrice() {
        double baseValue;
        double discountedValue;

        baseValue = orderLine.calculatePrice();
        discountedValue = customer.getDiscountedValue(this);
    }
}
```

```
    return discountedValue;  
}
```

```
public double getBaseValue() {  
    return orderLine.calculatePrice();  
}  
}
```

```
class OrderLine {  
    private Product product;  
    private int quantity;  
  
    public double calculatePrice() {  
        return product.getPrice(quantity);  
    }  
}
```

```
class Product {  
    public double getPrice(int quantity) {  
        // calculate price by quantity  
        return 0;  
    }  
}
```

```
class Customer {  
    public double getDiscountedValue(Order order) {  
        double baseValue = order.getBaseValue();
```

```
// calculate discount  
return baseValue * 0.9;  
}  
}
```